Zhihao Feng

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PROFESSIONAL SUMMARY

Founding Engineer at a stealth AI-for-science startup, architecting a closed-loop, self-evolving agentic research team and driving customer acquisition from prospect research to closed deals. Computational researcher building physics-grounded ML and simulation workflows spanning DFT, TDDFT, atomistic MD, and coarse-grained MD, especially for polymer semiconductors and sustainable polymers. Developed a physics-informed PRISM surrogate from 370+ GB CGMD data (∼400 systems; SANS-validated; > 90% benchmark wins; 60% less training data) and built *pyForceTune* to optimize force fields against DFT reference data (46% lower binding-energy error; < 0.1 kcal/mol dihedral error). Named a 2026 ACS PMSE Emerging Professionals Scholar (1 of 15 selected).

EDUCATION

Georgia Institute of Technology Atlanta, GA — 2021 – 2025
Ph.D. in Chemical Engineering (Minor: Data Science for the Chemical Industry)
Advisors: Prof. Seung Soon Jang and Prof. Natalie Stingelin

Cornell University Ithaca, NY — 2019 – 2021
M.S. in Chemical Engineering
Advisors: Prof. Rong Yang and Prof. Jeffrey D. Varner

University at Buffalo, SUNY Buffalo, NY — 2017 – 2019
B.S. in Chemical Engineering and B.A. in Chemistry (*Summa Cum Laude*)
Advisor: Prof. Haiqing Lin

PROFESSIONAL EXPERIENCE

Stealth Startup San Francisco, CA — Apr 2026 – Present
Founding Engineer

RESEARCH EXPERIENCE

Ph.D.: Polymer data/representation pipelines, DFT/MD simulation, and ML

- **ML-ready CGMD descriptors and physics-informed PRISM surrogate for polymer structure and property prediction (CGMD; ML; PRISM):** Developed an automated CGMD data/descriptor pipeline (**370+ GB**, **∼400** systems) and trained a physics-informed ML closure (*TensorFlow*) to predict polymer correlation functions and properties; outperformed standard analytical closures in **> 90%** of tests with **60%** less training data, SANS-validated, and deployed an interactive tool for rapid inference in minutes vs. MD in hours to days (***JCP***)
- **Force-field optimization toolkit for reliable conjugated polymer simulations (DFT; MM; MD; MLIP):** Developed ***pyForceTune***, a Python (*Pyomo*) toolkit to optimize bonded and nonbonded parameters against DFT reference data (*Jaguar*), validated against experimental solid-state structure; reduced binding-energy errors by **46%** vs. general force fields and achieved **< 0.1 kcal/mol** dihedral error vs. DFT (***SPIE Proc., 2025***); also explored fine-tuning UMA- and MACE-class ML interatomic potentials (*MatterTune*) as a complementary route to accurate atomistic simulation
- **Simulation-to-descriptor workflows tied to experiments (DFT; MD; Debye scattering):** Simulated Raman and IR spectra with DFT (including screening candidate degradation products against measured spectra), performed conformational sampling, and computed Debye scattering from MD trajectories across **> 30** conjugated polymer/small-molecule systems; extracted conformation/packing signatures to connect simulations to thin-film experimental observables (***Joule***, ***Adv. Mater.***, ***ACS Nano***, ***Mater. Horiz.***, ***Solar RRL***; **in review:** ***PNAS***; **in prep:** ***Nat. Energy***)
- **Large-scale polymer simulation for thermodynamics (MD; Debye scattering):** Ran repeated all-atom heating–cooling cycles in *LAMMPS* (**> 400 ns**) across temperature schedules; calculated order/alignment parameters and absolute entropy (2PT); identified drivers of assembly and thermal transitions to guide processing/stability decisions (**in prep**)
- **Conformation-ensemble electronic disorder analysis (DFT; MD):** Combined MD conformational sampling with DFT electronic-structure calculations to resolve static vs. sub-picosecond dynamic electronic disorder driven by low-frequency torsional modes and identify structural motifs that control disorder trends (**in revision:** ***Nat. Chem.***)

- **Electronic-structure features linked to polymer conformations (TDDFT; NTO):** Conducted TDDFT/NTO analysis across conjugated polymer, small-molecule acceptor, and polyurethane systems; connected structural changes to excited-state character to support stability and optical-property interpretation (***Prog. Org. Coatings***)
- **ML-driven screening under limited compute (MD/GCMC; SMILES/RDKit; Bayesian optimization):** Built a fast-MD + GCMC pipeline with refined force-field parameters to generate a **1,000**-polymer polyamine library with simulated CO_2 uptake (BRICS/RECAP fragment recombination; polyBERT/PolyNC embeddings); demonstrated budget-aware BO discovery ($N_0 = 10$, **budget = 30**; **20** retrospective trials) in which polymer-specific pretrained embeddings accelerated discovery of top sorbents (**in prep: *EES*; *AI4Mat-NeurIPS*, 2025**)
- **Mechanical response modeling (MD):** Modeled stress–strain behavior of conjugated polymers; quantified how chemical design and entanglement influence stretchability and toughness
- **Interface modeling for sustainable materials (MD):** Simulated PDMS–water and PDMS–octane interfaces; quantified surface tension and interfacial ordering to inform multi-surface repellency design

M.S.: ML-guided polymer discovery

- **Virtual screening for antibiofilm polymers (SMILES/RDKit; ML):** Built a virtual screening pipeline for **2,000+** amphiphilic copolymers using SMILES and RDKit-derived features and trained an ensemble model combining SVR with autoencoder-derived latent representations; identified and experimentally validated high-performance antibiofilm polymers (***Adv. Mater. Technol.*, 2023**).

COMPUTATIONAL SKILLS

Agentic AI:	Multi-agent LLM systems (Claude, OpenAI), closed-loop experiment orchestration, agent harness engineering, structured group debate, knowledge graphs, LLM tool use/MCP
Languages:	Python, R, Bash, Git
Key libraries:	TensorFlow, RDKit, Pyomo, SciPy, MDAnalysis, Two-Phase Thermodynamics, MatterTune
Software:	Schrödinger (Maestro, Jaguar), Gaussian, VASP, Materials Studio, LAMMPS
Infrastructure:	HPC/Slurm, GPU computing

PUBLICATIONS AND MANUSCRIPTS

Impact factors: Journal Citation Reports (Clarivate), 2025 release.

Peer-reviewed articles

[1] Pang Wang, **Zhihao Feng**, Florian Zimmermann, Seren Dilara Öz, Chen Chen, Wei Li, Ahmed Kadid, Maximilian Schiffer, Naho Kurahashi, Huiyue Ping, Yang Liu, Daan Coenen, Guorui He, Timo Maschwitz, Christian Tückmantel, Felix Lang, Wouter Maes, Yinhua Zhou, Tao Wang, Selina Olthof, Seung Soon Jang, Kai O. Brinkmann, and Thomas Riedl. "Disentangling Degradation of Non-Fullerene Organic Solar Cells Operated in Air." **Co-first author (corresponding author);** ***Joule (IF = 37.1), 2026, 11, 102681***

[2] **Zhihao Feng**, Christian T. Randolph, Tyler B. Martin, and Thomas E. Gartner III. "A Machine Learning Closure for Polymer Integral Equation Theory." ***Journal of Chemical Physics (IF = 3.7), 2026, 165, 164904***

[3] Min Gyu Kang, Rose Newman, **Zhihao Feng**, Jae Hoon Son, Charles Jeffreys, Jin-Woo Lee, Lingyun Zhao, Stanley Cazaly, Flurin Eisner, Byung Hui Lee, Min Hun Jee, Dong Won Kang, Martin Heeney, Tracey M. Clarke, Seung Soon Jang, James R. Durrant, and Han Young Woo. "Programming Intraparticle Ordering in Y6-Derived Organic Photocatalyst Nanoparticles for Efficient Solar Hydrogen Evolution." **Co-first author;** ***ACS Nano (IF = 17.3), 2026, 20, 22138***

[4] Hongmo Li, Henry J. Kantrow, David A. Valverde-Chávez, Meghan McNeil, Arianna Magni, Mohammad Balooch Qarai, Jude Kpare, Jaden Cramlet, Qiao He, Félix Thouin, **Zhihao Feng**, Yadong Zhang, Andrew Comstock, Stephen Barlow, Jason Azoulay, Seung Soon Jang, Renaud Demadrille, Martin Heeney, Seth Marder, Nicholas J. Hestand, Alberto Salleo, Frank C. Spano, Carlos Silva-Acuña, and Natalie Stingelin. "Trions as Fundamental Species in Chemically Doped Polymer Semiconductors." ***Advanced Materials (IF = 29.1), 2026, e00019***

[5] Jae Hoon Son, Keren Ai, Jaehyeong Kim, **Zhihao Feng**, Rose Newman, Yeonjeong Lee, Stanley Cazaly, Jongmin Han, Myoung Hoon Song, Flurin Eisner, Seung Soon Jang, Jin Young Kim, James R. Durrant, and Han Young Woo. "Redefining molecular design and exciton dynamics in single-component organic photocatalysts for efficient solar-to-hydrogen conversion." **Co-first author;** ***Materials Horizons (IF = 11.4), 2026, 13, 2918***

[6] Md Morshedur Rahman, Sehun Kim, Sung Hyun Kwon, **Zhihao Feng**, Nazrul Hsan, Seunga Choi, Ingi Hong, Chang Min Lee, Seung Soon Jang, and Joonseok Koh. "TD-DFT-guided reactive chromophore chain extension

for non-leaching neutral-black waterborne polyurethane coatings on nylon." ***Progress in Organic Coatings (IF = 7.9), 2026, 219, 110386***

[7] Minse Kim, Md Morshedur Rahman, Sung Hyun Kwon, **Zhihao Feng**, Nazrul Hsan, Ingi Hong, Seung Soon Jang, and Joonseok Koh. "Synthesis and properties of colored waterborne polyurethanes containing a uracil azo moiety: Experimental and computational study." ***Progress in Organic Coatings (IF = 7.9) 2025, 211, 109751***

[8] Maria G. D. Guaita, Rodrigo Szostak, Francisco M. C. da Silva, **Zhihao Feng**, Lucas Scalon, Verônica C. Teixeira, Tim Kodalle, Carolin M. Sutter-Fella, Seung Soon Jang, Hélio C. N. Tolentino, and Ana F. Nogueira. "Revealing the Crystallization Pathways of Mixed-Halide Low-Dimensional Perovskites: A First Step Toward Solar Cell Applications." ***Solar RRL (IF = 4.8), 2025, 9:e2500404***

[9] **Zhihao Feng**, Yifan Cheng, Alexandra Khlyustova, Aasim Wani, Trevor Franklin, Jeffrey D. Varner, Andrew L. Hook, and Rong Yang. "Virtual High-Throughput Screening of Vapor-Deposited Amphiphilic Polymers for Inhibiting Biofilm Formation." ***Advanced Materials Technologies (IF = 6.0) 2023, 8, 13, 2201533***

[10] Xiaoyi Chen, **Zhihao Feng**, Janavi Gohil, Christopher M. Stafford, Ning Dai, Liang Huang, and Haiqing Lin. "Reduced Holey Graphene Oxide Membranes for Desalination with Improved Water Permeance." ***ACS Applied Materials and Interfaces (IF = 7.8) 2020, 12, 1, 1387–1394***

Conference proceedings

[11] **Zhihao Feng**, Natalie Stingelin, and Seung Soon Jang. "Thermal behavior of semiconducting polymer assembly using multiscale modeling approach." ***Proc. SPIE, Organic, Hybrid, and Perovskite Photovoltaics XXVI, 2025***

Manuscripts submitted or in review/revision

[12] Henry J. Kantrow, Hongmo Li, Elizabeth Gutiérrez-Meza, **Zhihao Feng**, Spencer Yeager, Yael Tsarfati, Karen C. Bustillo, Peter Ercius, Xabier Rodríguez Martinez, Xingyuan Shi, Christina Cheng, Anna Kyrri, Luke Balhorn, Félix Thouin, Hao Li, Antonio Facchetti, Jenny Nelson, Jaime Martín, Qiao He, Martin Heeney, Sophia C. Hayes, Renaud Demadrille, Alberto Salleo, Sergei Tretiak, Erin L. Ratcliff, Seung Soon Jang, Carlos Silva-Acuña, and Natalie Stingelin. "Multiscale structure governs the disorder landscape in polymer semiconductors" **In revision at *Nature Chemistry (IF = 24.5)***

[13] Hongmo Li, Henry J. Kantrow, David Valverde-Chávez, Olivier Bardagot, **Zhihao Feng**, Jaden Cramlet, Sebastian Toyotoshi, Cyril Aumaitre, Antoine Curé, Seung Soon Jang, Alberto Salleo, Erin Ratcliff, Stephen Barlow, Renaud Demadrille, Seth R. Marder, Carlos Silva, and Natalie Stingelin. "Using vitrification to predict molecular limits for conjugated polymer:dopant interactions." **In review at *PNAS (IF = 9.5)***

[14] Indunil Angunawela, Subhrangsu Mukherjee, Somayeh Kashani, **Zhihao Feng**, Seung Soon Jang, and Harald Ade. "Mesophase ordering in ABA or ABA′ transverse amphiphilic semiconducting polymers." **In review at *Advanced Functional Materials (IF = 19.9)***

[15] **Zhihao Feng**, Matteo Palesati, Henry Kantrow, Jinwon Cho, Junhe Chen, Sung Hyun Kwon, Harald Ade, Jaime Martín, Natalie Stingelin, and Seung Soon Jang. "Beyond Backbone Planarity: Side-Chain Packing Governs Structural Order in Semiconducting Polymers." **Submitted to *Chemistry of Materials*, 2026**; related proceedings: ***SPIE Proc.*, 2025**

Manuscripts in preparation

[16] **Zhihao Feng**, Henry J. Kantrow, Mark D. Weber, Natalie Stingelin, and Seung Soon Jang. "Molecular Origin of Mesophase Transitions in pBTTT Revealed by Multiscale Modeling." **In preparation**

[17] Junhe Chen, Alif Qayyum, Nafiz Abeer, **Zhihao Feng**, Hyun Myung Woo, Seung Soon Jang, and Byung Jun Yoon. "Enabling Machine Learning-Assisted Discovery of Polyamines for Solid-State CO_2 Capture." **Co-first author; In preparation for *Energy and Environmental Science (IF = 30.5)*; related proceedings: *AI4Mat-NeurIPS*, 2025**

[18] Pang Wang, Kai O. Brinkmann, **Zhihao Feng**, Seung Soon Jang, and Thomas Riedl. "Understanding the Stability Limits of PM6:Y6 Organic Solar Cells under UV Irradiation and Nitrogen Environment." **In preparation for *Nature Energy (IF = 70.1)***

AWARDS AND HONORS

- 2026: PMSE Emerging Professionals Scholar — 1 of 15 selected, ACS
- 2025: Computational Allocation Award, National Renewable Energy Laboratory
- 2025: Amyn S. Teja International Research Endowment Fund, Georgia Tech
- 2024: Best Poster Award, Center for Organic Photonics and Electronics, Georgia Tech
- 2024: Beamtime Allocation Award, Paul Scherrer Institute
- 2024: Travel Award, College of Engineering, Georgia Tech
- 2023: Poster Runner-Up, Student Polymer Network, Georgia Tech
- 2022: Ph.D. Student Exemplary Achievement Award, Georgia Tech
- 2019: Dean's Achievement Award, University at Buffalo
- 2019: Academic Excellence in Chemistry Award, University at Buffalo
- 2019: Travel Award, North American Membrane Society
- 2019: Scholarship, School of Engineering and Applied Sciences, University at Buffalo
- 2019: Dean's List, University at Buffalo
- 2018: Dean's List, University at Buffalo

PROFESSIONAL SERVICES

- 2024–2026: Student Leader, Designing Materials to Revolutionize and Engineer our Future (DMREF) Program, Georgia Tech, NC State, and NIST
- 2024–2026: Creator, DMREF Repository and Portfolio Website, Georgia Tech
- 2024–2025: Maintainer, Natalie Stingelin Group Website, Georgia Tech
- 2024–2025: Member, SPIE Student Chapter, Georgia Tech
- 2022–2023: Member, School of Chemical & Biomolecular Engineering Safety Council, Georgia Tech

CONFERENCE PRESENTATIONS

[1] Zhihao Feng, Natalie Stingelin, and Seung Soon Jang. Presentation (invited), "Deciphering side-chain-functionalized polymer semiconductors: Insights on structure and thermal phase behavior from multiscale modeling." American Chemical Society Fall Meeting, Chicago, Illinois, USA, 2026.

[2] Zhihao Feng, Natalie Stingelin, and Seung Soon Jang. Presentation (contributed), "Thermal behavior of semiconducting polymer assembly using multiscale modeling approach." Society of Photo-Optical Instrumentation Engineers Optics + Photonics, San Diego, California, USA, 2025. Proceedings: *Proc. SPIE, Organic, Hybrid, and Perovskite Photovoltaics XXVI*.

[3] Zhihao Feng, Henry J. Kantrow, Natalie Stingelin, and Seung Soon Jang. Poster, "Unraveling the Influence of Side-Chain Design on Semiconducting Polymer Assembly: Insights from Multiscale Modeling." International Conference on Organic Electronics, Coimbra, Portugal, 2025.

[4] Zhihao Feng, Henry J. Kantrow, Mark Weber, Jinwon Cho, Junhe Chen, Omar Allam, Jiil Choi, Tyler Martin, Sung Hyun Kwon, Natalie Stingelin, and Seung Soon Jang. Presentation (contributed), "Unraveling the influence of side chain architectures on semiconducting polymer assembly: Insights from multiscale modeling." Fπ16 Symposium, Jeju, Korea, 2025.

[5] Zhihao Feng, Henry J. Kantrow, Jinwon Cho, Junhe Chen, Omar Allam, Sung Hyun Kwon, Jiil Choi, Natalie Stingelin, and Seung Soon Jang. Presentation, "Unraveling the influence of side chain architectures on semiconducting polymer assembly: Insights from multiscale modeling." American Chemical Society Fall Meeting, Denver, Colorado, USA, 2024.

[6] Zhihao Feng, Jinwon Cho, Junhe Chen, Omar Allam, and Seung Soon Jang. Presentation, "Investigating the effect of sidechains on self-assembly of semiconducting polymer using a multiscale modeling approach." American Physical Society March Meeting, Minneapolis, Minnesota, USA, 2024.

[7] Zhihao Feng and Thomas E. Gartner III. Poster, "Development of a machine learning-based closure relation for polymer integral equation theory." American Physical Society March Meeting, Las Vegas, Nevada, USA, 2023.

[8] Zhihao Feng, Yifan Cheng, Alexandra Khlyustova, Aasim Wani, Trevor Franklin, Jeffrey D. Varner, Andrew L. Hook, and Rong Yang. Presentation, "Virtual high-throughput screening of vapor-deposited amphiphilic polymers

for biofilm reduction with machine/deep learning." American Institute of Chemical Engineers Annual Meeting, Boston, MA, USA, 2021.